

Proof Sketch for Trust-Aware Fleet Estimation

Observer Law For Target j

Fix a target vehicle $j \in \mathcal{V}$. Host vehicle i runs $\text{DO}_i^{(j)}$ to estimate $x_j(t)$.

$$\hat{x}_{i,c}^{(j)}(t) = w_{ii}^{(j)}(t)\hat{x}_i^{(j)}(t) + w_{i0}^{(j)}(t)\hat{x}_0^{(j)}(t) + \sum_{l \in \mathcal{N}_i^{(j)}(t)} w_{il}^{(j)}(t)\hat{x}_l^{(j)}(t),$$

$$\hat{x}_i^{(j)}(t+1) = A_b \hat{x}_{i,c}^{(j)}(t) + B_b \hat{u}_j(t).$$

The residual self-weight is

$$w_{ii}^{(j)}(t) = 1 - w_{i0}^{(j)}(t) - \sum_{l \in \mathcal{N}_i^{(j)}(t)} w_{il}^{(j)}(t).$$

Interpretation

$w_{i0}^{(j)}$ anchors to LO_j , $w_{il}^{(j)}$ fuses trusted neighbor estimates, and $w_{ii}^{(j)}$ keeps confidence in host i 's own prediction.

From Trust Scores To Active Channels

For host vehicle i , the trust layer gives an opinion about all vehicles:

$$\mathcal{O}_i = [\mathcal{O}_i(1), \dots, \mathcal{O}_i(N)], \quad \mathcal{O}_i(l, t) \in [0, 1].$$

When $\text{DO}_i^{(j)}$ estimates target j , it uses only channels that are available and trusted enough:

$$\mathcal{N}_i^{(j)}(t) = \left\{ l \in \mathcal{N}_i : \hat{x}_l^{(j)}(t) \text{ is available, } \mathcal{O}_i(l, t) \geq \theta_{\min} \right\}.$$

Three channels used by the correction step

$$\underbrace{\hat{x}_0^{(j)}}_{\text{direct/local anchor}} \quad \underbrace{\{\hat{x}_l^{(j)}\}_{l \in \mathcal{N}_i^{(j)}(t)}}_{\text{trusted neighbors}} \quad \underbrace{\hat{x}_i^{(j)}}_{\text{host self-prediction}}.$$

If a channel becomes suspicious, the trust layer does not need to change the observer equation; it only gives that channel a smaller weight.

Raw Weight Scores Before Normalization

Choose nominal budgets

$$\bar{w}_0 > 0, \quad \bar{w}_{\text{self}} > 0, \quad \bar{w}_N = 1 - \bar{w}_0 - \bar{w}_{\text{self}} \geq 0.$$

Then assign nonnegative raw scores:

$$\tilde{w}_{i0}^{(j)}(t) = \bar{w}_0 \text{LT}_{i,j}(t) \phi_{i0}^{(j)}(t),$$

$$\tilde{w}_{ii}^{(j)}(t) = \bar{w}_{\text{self}} \phi_{ii}^{(j)}(t),$$

$$\tilde{w}_{il}^{(j)}(t) = \bar{w}_N \phi_{i,N}^{(j)}(t) \frac{\mathcal{O}_i(l, t)}{\sum_{p \in \mathcal{N}_i^{(j)}(t)} \mathcal{O}_i(p, t)}, \quad l \in \mathcal{N}_i^{(j)}(t).$$

Meaning

- $\text{LT}_{i,j}$ controls the direct/local anchor weight $w_{i0}^{(j)}$.
- $\mathcal{O}_i(l, t)$ distributes neighbor weight toward more trusted neighbors.
- $\phi_{i0}^{(j)}$, $\phi_{ii}^{(j)}$, $\phi_{i,N}^{(j)}$ are flag multipliers: they reduce or reinforce a channel depending on attack checks.

Normalization Gives Admissible Weights

Normalize the raw scores by their total mass:

$$Z_i^{(j)}(t) = \tilde{w}_{i0}^{(j)}(t) + \tilde{w}_{ii}^{(j)}(t) + \sum_{l \in \mathcal{N}_i^{(j)}(t)} \tilde{w}_{il}^{(j)}(t),$$

$$w_{iq}^{(j)}(t) = \frac{\tilde{w}_{iq}^{(j)}(t)}{Z_i^{(j)}(t)}, \quad q \in \{0, i\} \cup \mathcal{N}_i^{(j)}(t).$$

Why this is useful for the theorem

Since all raw scores are nonnegative,

$$\begin{aligned} w_{ii}^{(j)}(t) &\geq 0, & w_{i0}^{(j)}(t) &\geq 0, & w_{il}^{(j)}(t) &\geq 0, \\ w_{ii}^{(j)}(t) + w_{i0}^{(j)}(t) + \sum_{l \in \mathcal{N}_i^{(j)}(t)} w_{il}^{(j)}(t) &= 1. \end{aligned}$$

Hence

$$w_{i0}^{(j)}(t) + \sum_{l \in \mathcal{N}_i^{(j)}(t)} w_{il}^{(j)}(t) \leq 1,$$

which is the admissible-weight condition used in the ISS theorem.

Error Dynamics

Define, for fixed $j \in \mathcal{V}$,

$$\begin{aligned} e_0^{(j)}(t) &= \hat{x}_0^{(j)}(t) - x_j(t), & e_i^{(j)}(t) &= \hat{x}_i^{(j)}(t) - x_j(t), \\ e_{\text{DO}}^{(j)}(t) &= \text{col}\{e_1^{(j)}(t), \dots, e_N^{(j)}(t)\}. \end{aligned}$$

The local observer error is

$$e_0^{(j)}(t+1) = (A_b - F_j C_b) e_0^{(j)}(t) + d_0^{(j)}(t).$$

The distributed observer error is

$$\begin{aligned} e_{\text{DO}}^{(j)}(t+1) &= \left(w_0^{(j)}(t) \otimes A_b \right) e_0^{(j)}(t) \\ &\quad + \left((I_N - \tilde{\mathcal{L}}^{(j)}(t)) \otimes A_b \right) e_{\text{DO}}^{(j)}(t) + d_{\text{DO}}^{(j)}(t). \end{aligned}$$

Meaning Of The Row Sum

For row i of $I_N - \bar{\mathcal{L}}^{(j)}(t)$,

$$[I_N - \bar{\mathcal{L}}^{(j)}(t)]_{ii} = 1 - w_{i0}^{(j)}(t) - \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t), \quad [I_N - \bar{\mathcal{L}}^{(j)}(t)]_{il} = w_{il}^{(j)}(t).$$

Therefore,

$$\sum_{l=1}^N [I_N - \bar{\mathcal{L}}^{(j)}(t)]_{il} = 1 - w_{i0}^{(j)}(t) \leq 1.$$

Meaning of the row sum

$I_N - \bar{\mathcal{L}}^{(j)}(t)$ mixes only distributed-observer errors. The missing mass $w_{i0}^{(j)}(t)$ is not lost; it enters through the local-anchor input $(w_0^{(j)}(t) \otimes A_b)e_0^{(j)}(t)$.

If $w_{i0}^{(j)}(t) > 0$, the distributed-error mixing row is strictly less than one. This is why the local observer node 0 acts as an anchor in the ISS proof.

Sufficient ISS Conditions

Assume $\|d_0^{(j)}(t)\| \leq \bar{d}_0$ and $\|d_{\text{DO}}^{(j)}(t)\| \leq \bar{d}_{\text{DO}}$.

1) Local observer is internally stable.

$$A_{\ell,j} \triangleq A_b - F_j C_b \quad \text{is Schur stable.}$$

Meaning: if the local observer LO_j has no disturbance, its error $e_0^{(j)}$ decays to zero; with bounded disturbance, $e_0^{(j)}$ remains bounded.

2) Trust weights form an admissible correction.

$$w_{i0}^{(j)}(t) \geq 0, \quad w_{il}^{(j)}(t) \geq 0, \quad w_{i0}^{(j)}(t) + \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t) \leq 1.$$

Meaning: every received signal has nonnegative influence and the total external influence is capped; the remaining weight stays on the host prediction.

3) Distributed observer contracts at every switch.

$$\|(I_N - \tilde{\mathcal{L}}^{(j)}(t)) \otimes A_b\|_* \leq \alpha, \quad \alpha \in (0, 1).$$

Meaning: after each trust/topology update, the old distributed error is multiplied by a factor strictly smaller than one.

Then there exist $c_0, c_1 > 0$ and $\lambda \in (0, 1)$ such that

$$\begin{aligned}\|e_0^{(j)}(t)\| &\leq c_0 \lambda^t \|e_0^{(j)}(0)\| + c_0 \bar{d}_0, \\ \|e_{\text{DO}}^{(j)}(t)\| &\leq c_1 \lambda^t \|e_{\text{DO}}^{(j)}(0)\| + c_1 \left(\|e_0^{(j)}(0)\| + \bar{d}_0 + \bar{d}_{\text{DO}} \right).\end{aligned}$$

Reading the bound

Both errors are a decaying initial-error term plus bounded terms caused by local mismatch, model/input mismatch, and any bounded attack input absorbed into $d_{\text{DO}}^{(j)}$.

Proof Step 1: Local Observer Loop

Since $A_{\ell,j}$ is Schur, there exist $c_{\ell,j} > 0$ and $\lambda_{\ell,j} \in (0, 1)$ such that

$$\|A_{\ell,j}^t\|_* \leq c_{\ell,j} \lambda_{\ell,j}^t.$$

Unroll the local error recursion:

$$e_0^{(j)}(t) = A_{\ell,j}^t e_0^{(j)}(0) + \sum_{k=0}^{t-1} A_{\ell,j}^{t-1-k} d_0^{(j)}(k).$$

Using $\|d_0^{(j)}(k)\| \leq \bar{d}_0$,

$$\|e_0^{(j)}(t)\| \leq c_{\ell,j} \lambda_{\ell,j}^t \|e_0^{(j)}(0)\| + \frac{c_{\ell,j}}{1 - \lambda_{\ell,j}} \bar{d}_0.$$

Conclusion

The local anchor error is exponentially decaying up to a bounded disturbance term.

Proof Step 2: Distributed Observer Loop

Define

$$A_{\text{DO}}^{(j)}(t) = (I_N - \bar{\mathcal{L}}^{(j)}(t)) \otimes A_b, \quad B_0^{(j)}(t) = w_0^{(j)}(t) \otimes A_b.$$

The recursion becomes

$$e_{\text{DO}}^{(j)}(t+1) = A_{\text{DO}}^{(j)}(t)e_{\text{DO}}^{(j)}(t) + B_0^{(j)}(t)e_0^{(j)}(t) + d_{\text{DO}}^{(j)}(t).$$

Because the weights are nonnegative and normalized, $\|B_0^{(j)}(t)\|_* \leq \bar{B}_0$ for some $\bar{B}_0 > 0$. With the transition matrix $\Phi(t, k)$,

$$\|\Phi(t, k)\|_* \leq \alpha^{t-k}.$$

Thus,

$$\|e_{\text{DO}}^{(j)}(t)\| \leq \alpha^t \|e_{\text{DO}}^{(j)}(0)\| + \sum_{k=0}^{t-1} \alpha^{t-1-k} (\bar{B}_0 \|e_0^{(j)}(k)\| + \bar{d}_{\text{DO}}).$$

Insert the Step 1 bound and take

$$\lambda = \max\{\alpha, \lambda_{\ell,j}\} \in (0, 1).$$

The geometric sums are bounded, which gives the stated ISS bound.

How Attacks Enter The Theorem

The theorem is written for the stacked distributed error

$$e_{\text{DO}}^{(j)}(t+1) = A_{\text{DO}}^{(j)}(t)e_{\text{DO}}^{(j)}(t) + B_0^{(j)}(t)e_0^{(j)}(t) + d_{\text{DO}}^{(j)}(t), \quad (\text{T})$$

where $d_{\text{DO}}^{(j)}$ is only required to be bounded. It can collect model mismatch, input mismatch, noise, and attack-induced terms. If a received channel is corrupted before the correction step,

$$\tilde{x}_0^{(j)} = \hat{x}_0^{(j)} + a_0^{(j)}, \quad \tilde{x}_l^{(j)} = \hat{x}_l^{(j)} + a_l^{(j)},$$

then host i receives the additive attack input

$$d_{\text{att},i}^{(j)} = A_b \left(\chi_{i0}^{(j)} w_{i0}^{(j)} a_0^{(j)} + \sum_{l \in \mathcal{M}_i(t)} w_{il}^{(j)} a_l^{(j)} \right).$$

Thus the attacked observer has the same theorem form:

$$e_{\text{DO}}^{(j)}(t+1) = A_{\text{DO}}^{(j)} e_{\text{DO}}^{(j)} + B_0^{(j)} e_0^{(j)} + \tilde{d}_{\text{DO}}^{(j)}, \quad \tilde{d}_{\text{DO}}^{(j)} = d_{\text{DO}}^{(j)} + d_{\text{att}}^{(j)}.$$

Link to the theorem

No new stability proof is needed if $\tilde{d}_{\text{DO}}^{(j)}$ is bounded; the ISS bound is the same, with $\bar{d}_{\text{DO}}$ replaced by $\bar{\tilde{d}}_{\text{DO}}$.

Why Trust Gives Resilience

The attack input in row i is

$$d_{\text{att},i}^{(j)}(t) = A_b \left(\chi_{i0}^{(j)} w_{i0}^{(j)}(t) a_0^{(j)}(t) + \sum_{l \in \mathcal{M}_i(t)} w_{il}^{(j)}(t) a_l^{(j)}(t) \right).$$

If $\|a_0^{(j)}(t)\| \leq \bar{a}_0$ and $\|a_l^{(j)}(t)\| \leq \bar{a}_N$, then

$$\|d_{\text{att},i}^{(j)}(t)\| \leq \|A_b\|_* \left(\chi_{i0}^{(j)} w_{i0}^{(j)}(t) \bar{a}_0 + \sum_{l \in \mathcal{M}_i(t)} w_{il}^{(j)}(t) \bar{a}_N \right).$$

Key point

The attack effect is bounded by the malicious weight mass. Local-anchor attacks are attenuated by reducing $w_{i0}^{(j)}$ through $\text{LT}_{i,j}$ and $\phi_{i0}^{(j)}$; neighbor attacks are attenuated by reducing $w_{il}^{(j)}$ through $\mathcal{O}_i(l, t)$ or $\text{VT}_{i,l}$.